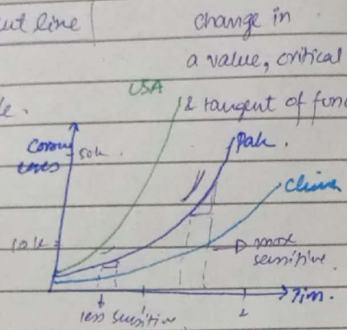


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Differentiation

Rate of change in dep variable
if we bring change in indep variable.

$y = f(x)$
dep. indep.
how fast/slow



Rate of change in dep variable wrt change in indep variable.

variable/quantity $\Rightarrow$ two values related $\rightarrow$ function.

change involved $\Rightarrow$ study of derivatives of functions

Theoretically

to study change in dep variable wrt indep variable's change.

"how much function is sensitive with its input."

covid cases function of USA is more sensitive as compared to China & Pak

Mathematically,

$$\frac{dC_{usa}}{dt} > \frac{dC_{pak}}{dt} > \frac{dC_{china}}{dt}$$

The derivative of a function represents how much a function is sensitive to its input.

"from graph no of covid cases func for USA is most sensitive to its input (compared to no of covid cases function for Pak & China)."

Similar for decreasing/declining nature.

Instantaneous $\rightarrow$ at a particular point.

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Avg = change in speed / change in time

40 km/s $t=5$ 60 km/s $t=10$

Mathematical Definition of Derivative.

$$y = f(x)$$

$$y + \delta y = f(x + \delta x)$$

change in dep variable.

change in input / indep variable.

$$\delta y = f(x + \delta x) - y$$

$$\Rightarrow \delta y = f(x + \delta x) - f(x)$$

$$\frac{\delta y}{\delta x} = \frac{f(x + \delta x) - f(x)}{\delta x}$$

average rate of change.

$\rightarrow$ some interval.

change in y wrt change in x (average change or average ratio)

To find instantaneous change,

$\Delta x \rightarrow 0$ Change in x is very very small. Indep variable is very small.

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\Rightarrow \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Instantaneous

tells instantaneous change in dep variable wrt to indep variable.

- Abinitio

- First Principle

- Definition of Derivative.

(slope of tangent line)

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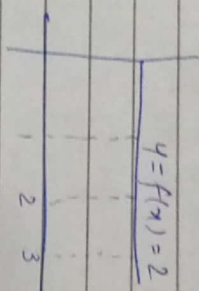
Techniques of Derivatives

$$\Rightarrow \frac{d}{dx} (\text{constant}) = 0 \quad \text{Newton}$$

$$f'(x)$$

$$D(f) \rightarrow D = \frac{d}{dx}$$

law



constant has zero

sensitivity

$$\frac{d}{dx} (x) = 1$$

$\frac{d}{dx} \rightarrow$ derivative with x , $\frac{d}{dt} \rightarrow$ derivative with t

no change
no sensitivity

$$\Rightarrow \frac{d}{dx} (2f(x)) = 2 \frac{d}{dx} (f(x))$$

$$\Rightarrow \frac{d}{dx} (k f(x)) = k \frac{d}{dx} (f(x))$$

constant rule

$$\Rightarrow \frac{d}{dx} (f(x) \pm g(x)) = \frac{d}{dx} f \pm \frac{d}{dx} g$$

Sum/difference rule

$$\Rightarrow \frac{d}{dx} (f(x) \cdot g(x)) = f(x) \frac{d}{dx} (g(x)) + g(x) \frac{d}{dx} (f(x))$$

$$\Rightarrow \frac{d}{dx} (f(x) \cdot g(x)) = f(x) \frac{d}{dx} g + g(x) \frac{d}{dx} f$$

$$\frac{d}{dx} (f(x) \cdot g(x) \cdot h(x)) = f(x) \frac{d}{dx} g + f(x) \cdot h(x) \cdot \frac{d}{dx} g + g(x) \frac{d}{dx} h$$

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$$\Rightarrow \frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \frac{d}{dx} f - f(x) \frac{d}{dx} g}{(g(x))^2}$$

Quotient Rule

$\Rightarrow$ Power Rule of Differentiation

Power function = (variable)

$$f(x) = x^a \rightarrow \text{const}$$

a can be $+$ ve or $-$ ve

$$y = 2^x$$

$$\frac{dy}{dx} \neq x \cdot 2^{x-1}$$

shift constant power to start & reduce power by one.
only applicable to power function.

$$\textcircled{1} f(x) = \frac{1}{x}$$

$$\frac{df}{dx} = -1 \cdot x^{-2} = -1 \cdot x^{-2} = -x^{-2}$$

$$\textcircled{2} f(x) = \sqrt{x} = x^{1/2}$$

$$\frac{df}{dx} = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}}$$

$$\textcircled{3} f(x) = \frac{1}{\sqrt[3]{x}} = x^{-1/3}$$

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Generalized Power Rule:-

$$f(x) = \text{function}^{\text{constant}}$$

$$\frac{df}{dx} = \text{constant} \cdot \frac{d(\text{function})}{dx}$$

① $f(x) = (2x^2 + x - 1)^{1/2} \Rightarrow \frac{df}{dx} = \frac{d}{dx} (2x^2 + x - 1)^{1/2}$

$$\frac{df}{dx} = \frac{1}{2} (2x^2 + x - 1)^{-1/2} \cdot \frac{d}{dx} (2x^2 + x - 1)$$

$$= \frac{1}{2 \sqrt{2x^2 + x - 1}} (4x + 1 - 0)$$

$$= \frac{4x + 1}{2 \sqrt{2x^2 + x - 1}}$$

$$= \frac{1}{2} (2x^2 + x - 1)^{-1/2} \left[\frac{d}{dx} (2x^2) + \frac{d}{dx} (x) - \frac{d}{dx} (1) \right]$$

$$= \frac{1}{2} (2x^2 + x - 1)^{-1/2} [2 \cdot 2x + 1]$$

② $f(x) = \sqrt[3]{\frac{2x^2 + 1}{x - 1}}$

First apply power rule then Quotient.

$$f(x) = \left(\frac{2x^2 + 1}{x - 1} \right)^{1/3}$$

$$\frac{df}{dx} = \frac{1}{3} \left(\frac{2x^2 + 1}{x - 1} \right)^{-2/3} \cdot \frac{d}{dx} \left(\frac{2x^2 + 1}{x - 1} \right)$$

$$= \frac{1}{3} \left(\frac{2x^2 + 1}{x - 1} \right)^{-2/3} \left[\frac{(x - 1) \frac{d}{dx} (2x^2 + 1) - (2x^2 + 1) \frac{d}{dx} (x - 1)}{(x - 1)^2} \right]$$

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$$= \frac{1}{3} \left(\frac{2x^2 + 1}{x - 1} \right)^{-2/3} \cdot \frac{(x - 1)(4x) - (2x^2 + 1)(1)}{(x - 1)^2}$$

$$= \frac{1}{3} \left(\frac{2x^2 + 1}{x - 1} \right)^{-2/3} \cdot \frac{4x(x - 1) - (2x^2 + 1)}{(x - 1)^2}$$

$$= \frac{1}{3} \left(\frac{2x^2 + 1}{x - 1} \right)^{-2/3} \cdot \frac{4x^2 - 4x - 2x^2 - 1}{(x - 1)^2}$$

$$= \frac{1}{3} \cdot \frac{(x - 1)(2x^2 - 4x - 1)}{(2x^2 + 1)^{2/3} \cdot (x - 1)^2} \cdot \frac{df}{dx} = \frac{1}{3} \cdot \frac{(x - 1)}{(2x^2 + 1)^{2/3}} \cdot \frac{(2x^2 - 4x - 1)}{(x - 1)^2}$$

$$\frac{df}{dx} = \frac{2x^2 - 4x - 1}{3(2x^2 + 1)^{2/3} (x - 1)}$$

Ex: $f(x) =$

Derivative of Exponential function:-

Exponential Rule.

Exponential = (constant)^{variable}
function

$$y = a^x$$

only positive, shouldn't be 1.

$y = 2^x$ Computers.
 $y = 10^x$ Physics.
 $y = e^x$ Maths.

$$y = 5^x$$

$$y = \left(\frac{1}{2}\right)^x$$

$$\ln e = 1$$

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$$\Rightarrow \frac{d}{dx} (c \cdot f) = \text{constant} \cdot \frac{d}{dx} (\text{variable})$$

$$\frac{d}{dx} (c \cdot f) = \text{constant} \cdot \ln(\text{constant})$$

$$y = 2^x$$

$$\frac{dy}{dx} = 2^x \cdot \ln 2$$

$$y = a^x$$

$$\frac{dy}{dx} = a^x \cdot \ln a$$

$$y = e^x$$

$$\frac{dy}{dx} = e^x \cdot \ln e$$

$$f(x) = 2^{x^2+1}$$

$$\frac{df}{dx} = 2^{x^2+1} \cdot \ln 2 \cdot \frac{d}{dx} (x^2+1)$$

Generalized Exponential function Rule.

$$f(x) = (\text{constant})^{\text{function}}$$

$$\frac{df}{dx} = (\text{constant})^{\text{function}} \cdot \ln(\text{constant}) \cdot \frac{d}{dx} (\text{function})$$

Derivative of log function

$$y = \log_a x$$

inverse of exponential function. $\Rightarrow$

$$f(x) = 2^x$$

$$f^{-1}(x) = \log_2 x$$

$$\frac{dy}{dx} = \frac{1}{x \ln a}$$

$$f(x) = 10^x$$

$$f^{-1}(x) = \log_{10} x$$

$$= \log x / \log 10$$

common log.

$$f(x) = 3^x$$

$$f^{-1}(x) = \log_3 x$$

$$f(x) = e^x$$

$$f^{-1}(x) = \log_e x = \ln x$$

natural log.

$$\frac{d}{dx} \left(\frac{1}{\cos x} \right) = \frac{d}{dx} (\cos x)^{-1} = -1(\cos x)^{-2} \cdot -\sin x = \frac{\sin x}{\cos^2 x} = \tan x \cdot \sec x$$

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$$y = \log_2 x$$

$$y = \ln x$$

$$\frac{dy}{dx} = \frac{1}{x \ln 2}$$

$$\frac{dy}{dx} = \frac{1}{x \ln e} = \frac{1}{x}$$

$$y = \ln(x^2+1)$$

$$\frac{dy}{dx} = \frac{1}{x^2+1} \cdot \frac{1}{\ln e} \cdot \frac{d}{dx} (x^2+1)$$

$$\Rightarrow \frac{d}{dx} (\log_a (\text{angle}))$$

$$= \frac{1}{\ln a} \cdot \frac{1}{\text{angle}} \cdot \frac{d}{dx} (\text{angle})$$

Derivatives of Trigonometric functions.

$$\frac{d}{dx} (\sin x) = \cos x$$

$$\frac{d}{dx} (\sec x) = \sec x \tan x$$

Sin - cos

Tan - sec

cot - cosec

$$\frac{d}{dx} (\cos x) = -\sin x$$

$$\frac{d}{dx} (\csc x) = -\csc x \cot x$$

derivative of all the functions that start from c are -ve.

$$\frac{d}{dx} (\tan x) = \sec^2 x$$

$$\frac{d}{dx} (\cot x) = -\csc^2 x$$

can apply quotient rule i.e. $\frac{d}{dx} \left(\frac{\sin x}{\cos x} \right)$

derivative main func khud ya us ka pair ata hai.

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$$\frac{d(\csc x)}{dx} = \frac{d}{dx} \left(\frac{1}{\sin x} \right) = \frac{d(\sin x)^{-1}}{dx}$$

$$= -1(\sin x)^{-2} \cdot \cos x$$

$$= -\cos x$$

$$(\sin x)^2$$

$$= -\cos x \cdot \frac{1}{\sin x}$$

$$= -\cot x \cdot \csc x$$

$$\frac{d(\sin(\text{angle}))}{dx}$$

$$= \cos(\text{angle}) \cdot \frac{d(\text{angle})}{dx}$$

$$\Rightarrow \frac{d(\sec(\ln x))}{dx} = \sec(\ln x) \tan(\ln x) \cdot \frac{d(\ln x)}{dx}$$

$$= \sec(\ln x) \tan(\ln x) \cdot \frac{1}{x}$$

$$\Rightarrow \frac{d(\tan(2^x))}{dx} = \sec^2(2^x) \cdot \frac{d(2^x)}{dx}$$

$$= \sec^2(2^x) \cdot (2^x \cdot \ln 2 \cdot \frac{d(x^2)}{dx})$$

Derivatives of Inverse Trigonometric function (5-formulas)

$$y = \sin^{-1} x = \arcsin(x)$$

$$\sin^{-1}(\sin x) = x$$

cancel
each other

$$\sin(\sin^{-1} x) = x$$

$$y = \sin^{-1} x$$

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$$\frac{dy}{dx} = ?$$

(sin both sides)

$$\sin y = x$$

Diff wrt x

$$\cos y \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cos y}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-\sin^2 y}} \quad (\cos y = \sqrt{1-\sin^2 y})$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

Similar for $\tan^{-1} x$

$$y = \tan^{-1} x \Rightarrow \tan y = x$$

Diff wrt x

$$\sec^2 y \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$\sec^2 y + \tan^2 y = 1$$

$$= \frac{1}{1+\tan^2 y}$$

$$\frac{d(\tan^{-1} x)}{dx} = \frac{1}{1+x^2}$$

$$\frac{1}{\sin x} \neq \sin^{-1} x$$

Power Rule

$$f(x) = x^a = \frac{d}{dx} x^a = a x^{a-1}$$

$$f(x) = (g(x))^a$$

$$\frac{df}{dx} = a(g(x))^{a-1} \cdot \frac{dg}{dx}$$

Exponential Rule

$$f(x) = a^x = \frac{d}{dx} a^x = a^x \cdot \ln a$$

$$f(x) = a^{g(x)}$$

$$\frac{df}{dx} = a^{g(x)} \cdot \ln a \cdot \frac{d(g(x))}{dx}$$

Log function

$$y = \log_a(x) = \frac{dy}{dx} = \frac{1}{x \ln a}$$

$$y = \log(f(x))$$

$$\frac{dy}{dx} = \frac{1}{f(x)} \cdot \ln a \cdot \frac{d(f(x))}{dx}$$

Trigonometric functions

$$\frac{d(\sin(f(x)))}{dx} = \cos(f(x)) \cdot \frac{df}{dx}$$

$$\frac{d(\cos(f(x)))}{dx} = -\sin(f(x)) \cdot \frac{df}{dx}$$

$$\frac{d(\tan(f(x)))}{dx} = \sec^2(f(x)) \cdot \frac{df}{dx}$$

$$\frac{d(\cot(f(x)))}{dx} = -\csc^2(f(x)) \cdot \frac{df}{dx}$$

$$\frac{d(\sec(f(x)))}{dx} = \sec(f(x)) \tan(f(x)) \cdot \frac{df}{dx}$$

$$\frac{d(\csc(f(x)))}{dx} = -\csc(f(x)) \cot(f(x)) \cdot \frac{df}{dx}$$

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$$\frac{d}{dx} (\sin^{-1}(\text{angle})) = \frac{1}{\sqrt{1-(\text{angle})^2}} \cdot \frac{d}{dx} (\text{angle}) = -\frac{d}{dx} (\cos^{-1}(\text{angle}))$$

$$\frac{d}{dx} (\tan^{-1}(\text{angle})) = \frac{1}{1+(\text{angle})^2} \cdot \frac{d}{dx} (\text{angle}) = -\frac{d}{dx} (\cot^{-1}(\text{angle}))$$

$$\frac{d}{dx} (\sec^{-1}(\text{angle})) = \frac{1}{\text{angle} \sqrt{(\text{angle})^2 - 1}} \cdot \frac{d}{dx} (\text{angle}) = -\frac{d}{dx} (\csc^{-1}(\text{angle}))$$

$$\frac{d}{dx} (\cos^{-1}(\text{angle})) = -\frac{1}{\sqrt{1-(\text{angle})^2}}$$

$$\frac{d}{dx} (\cot^{-1}(\text{angle})) = -\frac{1}{1+(\text{angle})^2} \cdot \frac{d}{dx} (\text{angle})$$

$$\frac{d}{dx} (\csc^{-1}(\text{angle})) = -\frac{1}{\text{angle} \sqrt{(\text{angle})^2 - 1}} \cdot \frac{d}{dx} (\text{angle})$$

Exercise

Find $\frac{dy}{dx}$ if

$$y = \left(\ln \left(\sin^{-1} \left(\frac{2^x + \tan x}{x^2 + \sec x} \right) \right) \right)^{3/2}$$

Power
log
sin
Quotient

Differentiating w.r.t x

$$\frac{dy}{dx} = \frac{3}{2} \left(\ln \left(\sin^{-1} \left(\frac{2^x + \tan x}{x^2 + \sec x} \right) \right) \right)^{3/2 - 1} \cdot \frac{d}{dx} \left(\ln \left(\sin^{-1} \left(\frac{2^x + \tan x}{x^2 + \sec x} \right) \right) \right)$$

solve this now,

natural log = $\ln = \log_e$

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$$\frac{dy}{dx} = \frac{1}{\sin^{-1} \left(\frac{2^x + \tan x}{x^2 + \sec x} \right)} \cdot \frac{d}{dx} \left(\ln \left(\sin^{-1} \left(\frac{2^x + \tan x}{x^2 + \sec x} \right) \right) \right)$$

Now $\frac{d}{dx} \left(\sin^{-1} \left(\frac{2^x + \tan x}{x^2 + \sec x} \right) \right)$

$$= \frac{1}{\sqrt{1 - \left(\frac{2^x + \tan x}{x^2 + \sec x} \right)^2}} \cdot \frac{d}{dx} \left(\frac{2^x + \tan x}{x^2 + \sec x} \right)$$

$$= \frac{x^2 + \sec x}{\sqrt{(x^2 + \sec x)^2 - (2^x + \tan x)^2}} \cdot \left[\frac{(x^2 + \sec x) \frac{d}{dx} (2^x \ln 2 + \sec x) - (2^x + \tan x) \frac{d}{dx} (x^2 + \sec x)}{(x^2 + \sec x)^2} \right]$$

put in equation (1).

$$\frac{dy}{dx} = \frac{3}{2} \left(\ln \left(\sin^{-1} \left(\frac{2^x + \tan x}{x^2 + \sec x} \right) \right) \right)^{3/2 - 1} \cdot \frac{1}{\sin^{-1} \left(\frac{2^x + \tan x}{x^2 + \sec x} \right)} \cdot \frac{(x^2 + \sec x) \left[\ln 2 + \sec x \right] - (2^x + \tan x) (2x + \sec x \tan x)}{(x^2 + \sec x)^2}$$